

Mall Customer Segmentation

Data-Driven Insights using 3D K-Means Clustering

Prepared By: Palwasha Mushtaq

Month : 2

Internship Task : 3

Tools Used: Python, Jupyter Notebook,

Internship Domain: Data Science

Submitted To: Arch Technologies

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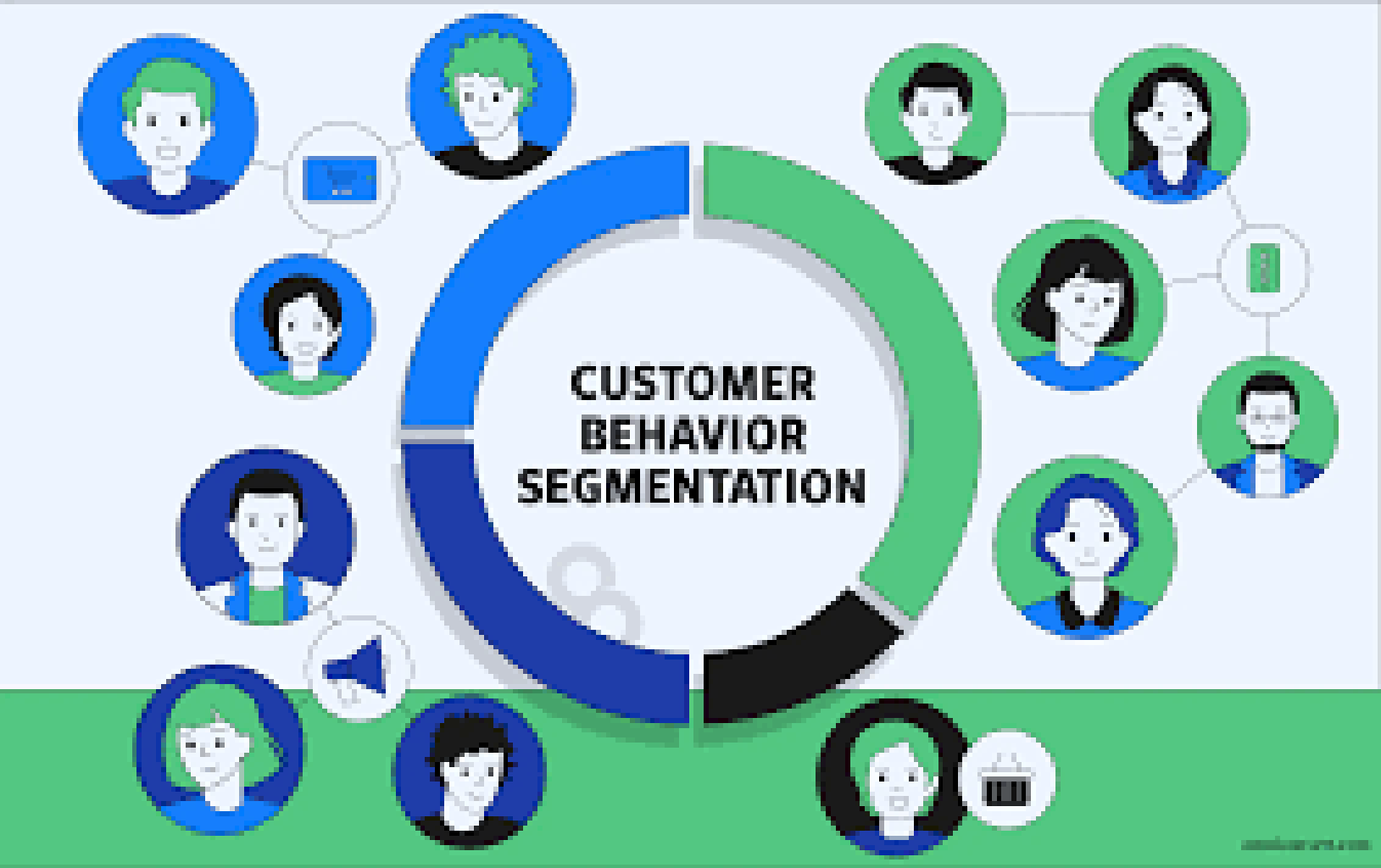
Introduction & Project Overview

- This project focuses on Customer Segmentation, which is the practice of dividing a company's customer base into distinct groups that reflect similarity among customers in each group. In the retail sector, understanding customer behavior is crucial for optimizing marketing strategies, improving customer retention, and maximizing revenue.
- The objective of this notebook is to analyze a dataset of mall customers and group them into distinct segments based on their demographic characteristics and purchasing data. By identifying these hidden patterns, the mall's marketing team can design targeted campaigns, offer personalized promotions, and improve the overall customer experience.



Business Problem & Objective

- **Problem:** The mall management wants to understand their customers better to target the right audience with specific marketing strategies (e.g., spending offers, loyalty programs).
- **Objective:** Group customers into different segments based on features like Age, Gender, Annual Income, and Spending Score using unsupervised machine learning techniques (like K-Means Clustering).



Data Source & Overview

Dataset Profile Summary

The mall customer dataset captures behavioral data of shoppers, allowing us to find patterns without preset labels.

- Rows: 200 consumer profiles
- Key Features: Age, Annual Income (\$k), Spending Score (1-100)
- Data Types: Clean numerical and categorical integer steps



```
DATA SHAPE:
(200, 5)

=====
DATA COLUMNS:
Index(['CustomerID', 'Gender', 'Age', 'Annual Income (k$)',
      'Spending Score (1-100)'],
      dtype='object')
=====
```

RangeIndex: 200 entries, 0 to 199

Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	CustomerID	200 non-null	int64
1	Gender	200 non-null	object
2	Age	200 non-null	int64
3	Annual Income (k\$)	200 non-null	int64
4	Spending Score (1-100)	200 non-null	int64

dtypes: int64(4), object(1)

DESCRIPTIVE STATISTIC SUMMARY:				
	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

```
=====
NO OF UNIQUE VALUES IN DATASET:
CustomerID      200
Gender           2
Age             51
Annual Income (k$)  64
Spending Score (1-100)  84
dtype: int64
```


Libraries Used & Preprocessing

Feature Standardization Strategy

Because Annual Income (measured in thousands) has a naturally wider range than Age, Scikit-Learn's StandardScaler was used.

Why? This step guarantees distance calculations in K-Means aren't biased towards higher scale features.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
import plotly.express as px
from sklearn.cluster import KMeans
```

LABEL ENCODING:

	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	19	15	39
1	1	21	15	81
2	0	28	16	6
3	0	23	16	77
4	0	31	17	40

• StandardScaler

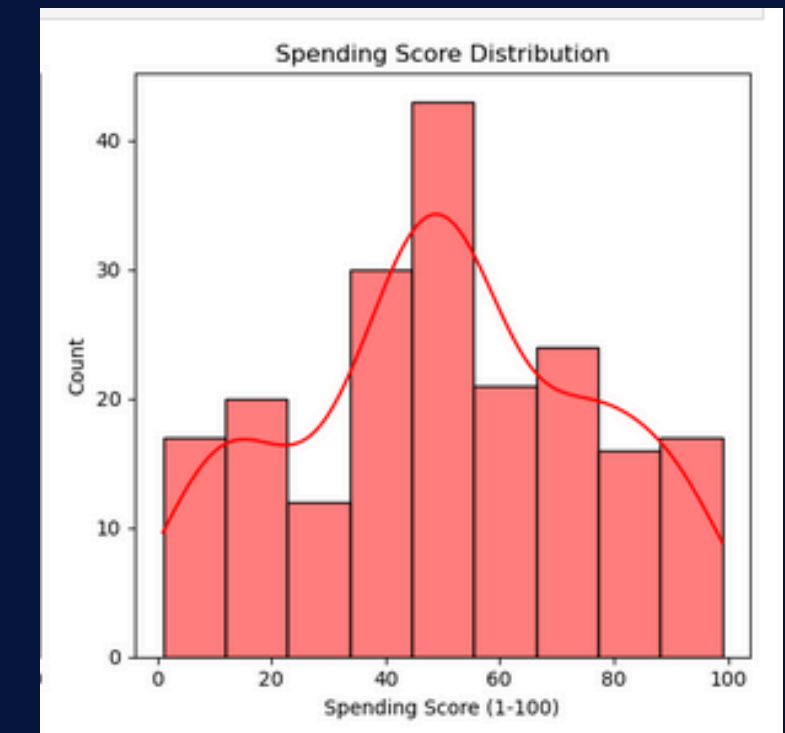
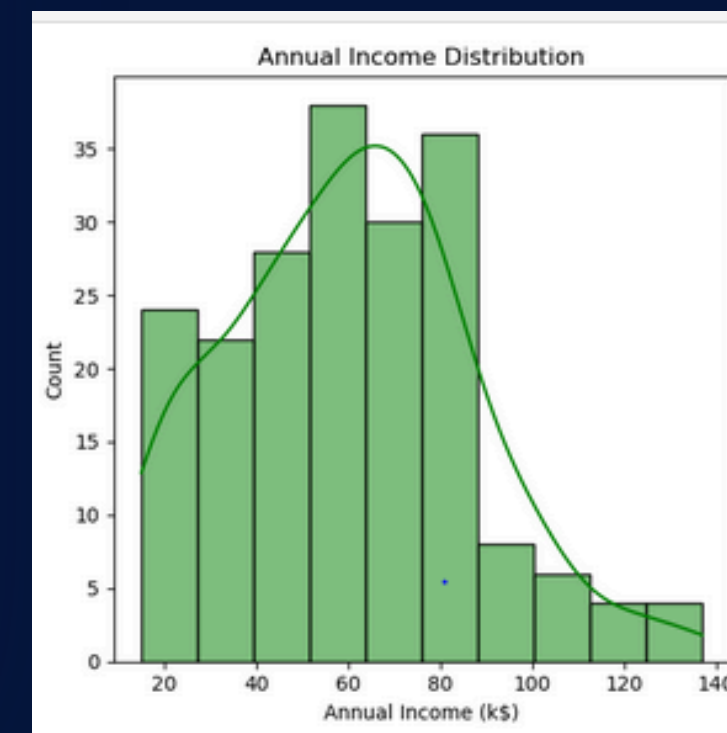
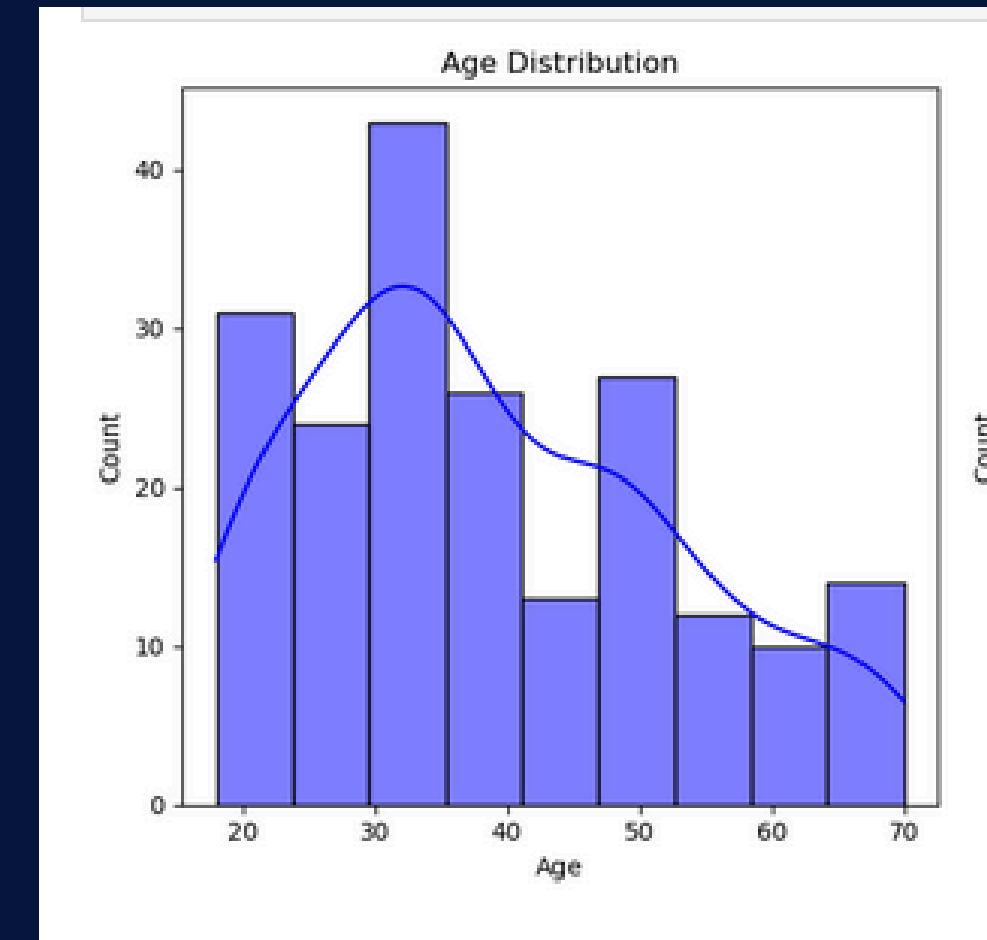
```
standardscaler = StandardScaler()
Col = ['Age', 'Annual Income (k$)', 'Spending Score (1-100)']
standardscaled = standardscaler.fit_transform(df[Col])
print('STANDARD SCALER: \n' , standardscaled)
```

STANDARD SCALER:

```
[[-1.42456879 -1.73899919 -0.43480148]
 [-1.28103541 -1.73899919  1.19570407]
 [-1.3528021  -1.70082976 -1.71591298]
 [-1.13750203 -1.70082976  1.04041783]
 [-0.56336851 -1.66266033 -0.39597992]
 [-1.20926872 -1.66266033  1.00159627]
 [-0.27630176 -1.62449001 -1.71591298]
 [-1.13750203 -1.62449001  1.70038436]
 [ 1.00493225 -1.58632148 -1.83237767]
 [-0.6351352  -1.58632148  0.84631002]]
```

Exploratory Data Analysis (EDA)

- **Age Distribution (Left):** The customer base is highly active between the ages of 30 and 40. There is a noticeable drop-off in customers older than 55, indicating the mall primarily attracts a younger to middle-aged demographic.
- **Annual Income Distribution (Center):** Most customers earn an annual income in the range of \$50k to \$85k. The data is slightly right-skewed, showing a smaller number of high-earning individuals making over \$100k.
- **Spending Score Distribution (Right):** The spending score shows a prominent peak right around the 40 to 60 mark. This tells us that the majority of customers exhibit average spending habits, while the rest are evenly distributed between low and high spenders.**



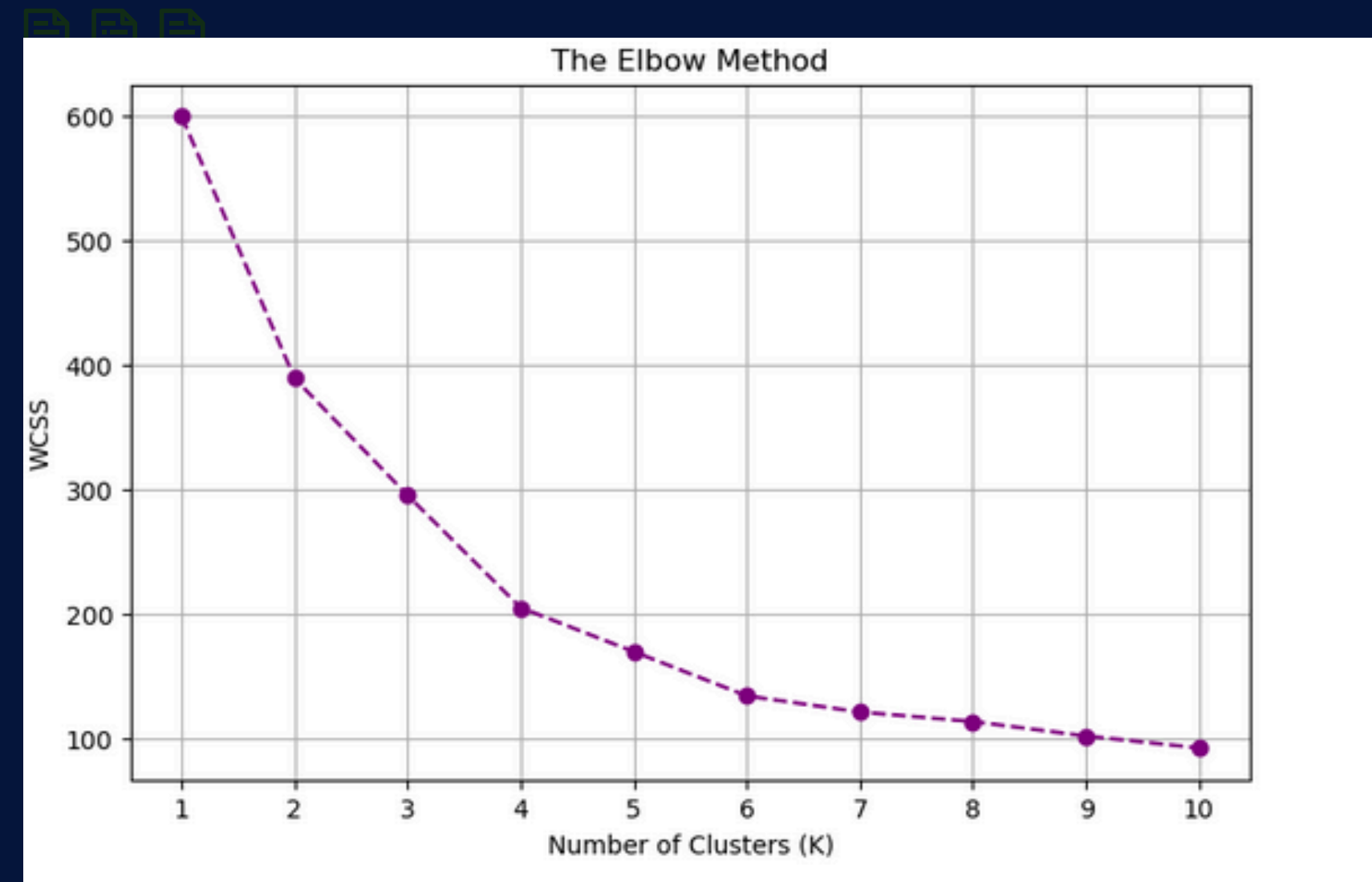
Finding Optimal Clusters: The Elbow Method

The Elbow Method

We ran the K-Means algorithm for K=1 to 10. The WCSS (Within-Cluster Sum of Squares) graph showed a clear "Elbow at K=5

Key Insight: 5 groups provide the most balanced mathematical separation for our customer base.

- **Elbow Method plots WCSS against the number of clusters (K).**
- **Sharp decrease in WCSS from K = 1 to K = 4 indicates better cluster formation.**
- **After K = 4 or K = 5, the curve starts flattening, showing minimal improvement.**
- **The "elbow point" appears around K = 5.**
- **K = 5 is selected as the optimal number of clusters for this 3D clustering dataset.**



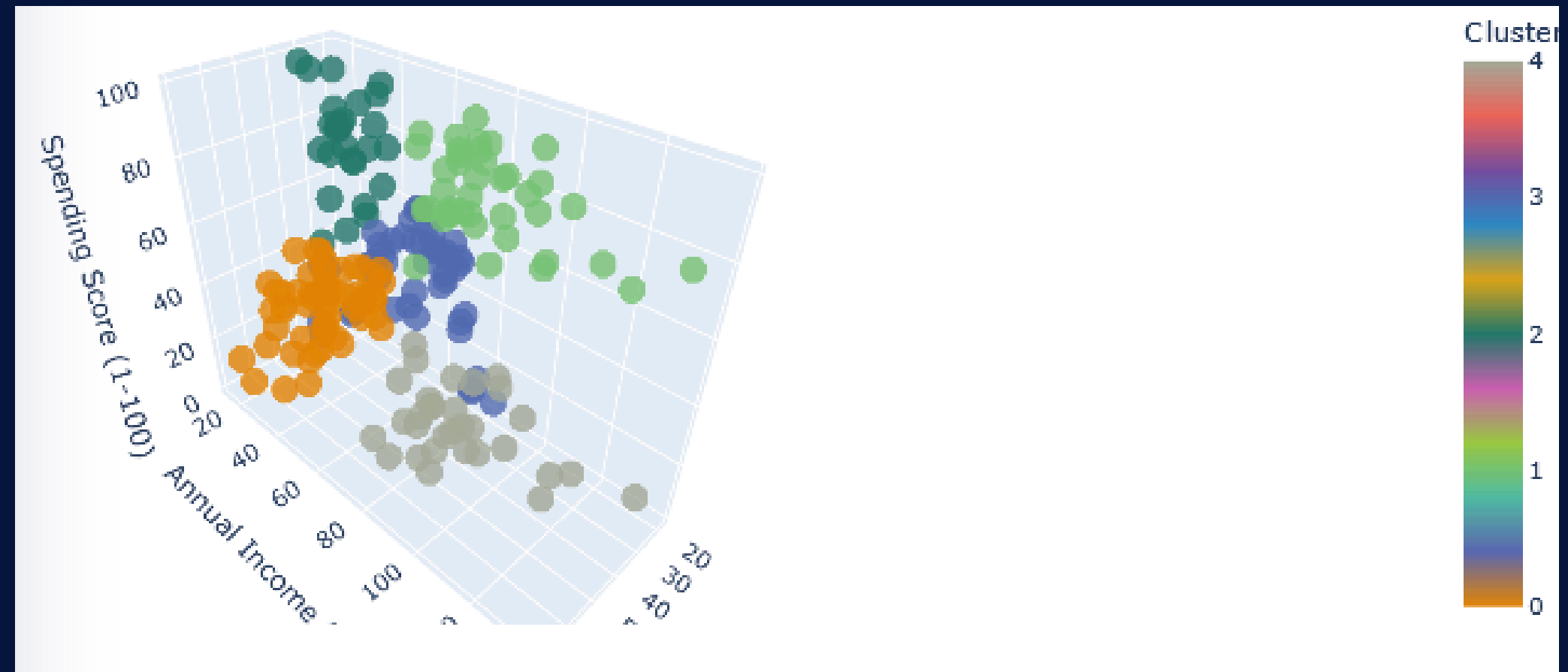
Train the K-Means Model

- The `df.head()` output confirms that a new **Cluster** column has been added successfully.
- K-Means classified each customer into one of the 5 clusters (0-4).
- Customers 0, 1, 3, and 4 are grouped into **Cluster 2** due to similar income and spending behavior.
- These customers have low income but moderate-to-high spending scores.
- Customer 2 is placed in Cluster 3 because of a very low spending score of 6.
- This shows that K-Means identifies distinct customer behavior patterns and creates clear cluster boundaries.

	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	Cluster
0	1	19	15	39	2
1	1	21	15	81	2
2	0	20	16	6	3
3	0	23	16	77	2
4	0	31	17	40	2

3D View of Mall Customer Segments

- This 3D scatter plot represents Mall Customer Segmentation using clustering.
- Customers are grouped into 5 different clusters based on Age, Annual Income, and Spending Score.
- Each color represents a separate customer group with similar purchasing behavior.
- The visualization helps identify high-spending, low-spending, and average customers.
- 3D clustering provides a clearer understanding of customer patterns and market segmentation.



Discovered Market Segments (K=5)



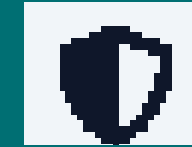
VIP Stars

High Income, High
Spending Score.
The primary
revenue drivers.



Carefree Young

Young demographic,
Low Income but High
Spending Score.



Conservative

High Income but
extremely low
Spending Scores.



Sensible Spenders

Middle-aged with
average income and
spending habits.



Budget Shoppers

Price-sensitive
customers with Low
Income and Low
Spend.

Persona: VIP Stars

- **Profile:** These are high-earning individuals who actively engage with the mall's offerings.
- **Strategy:** Exclusive VIP lounges, early access to new collections, and personal shopping assistants.

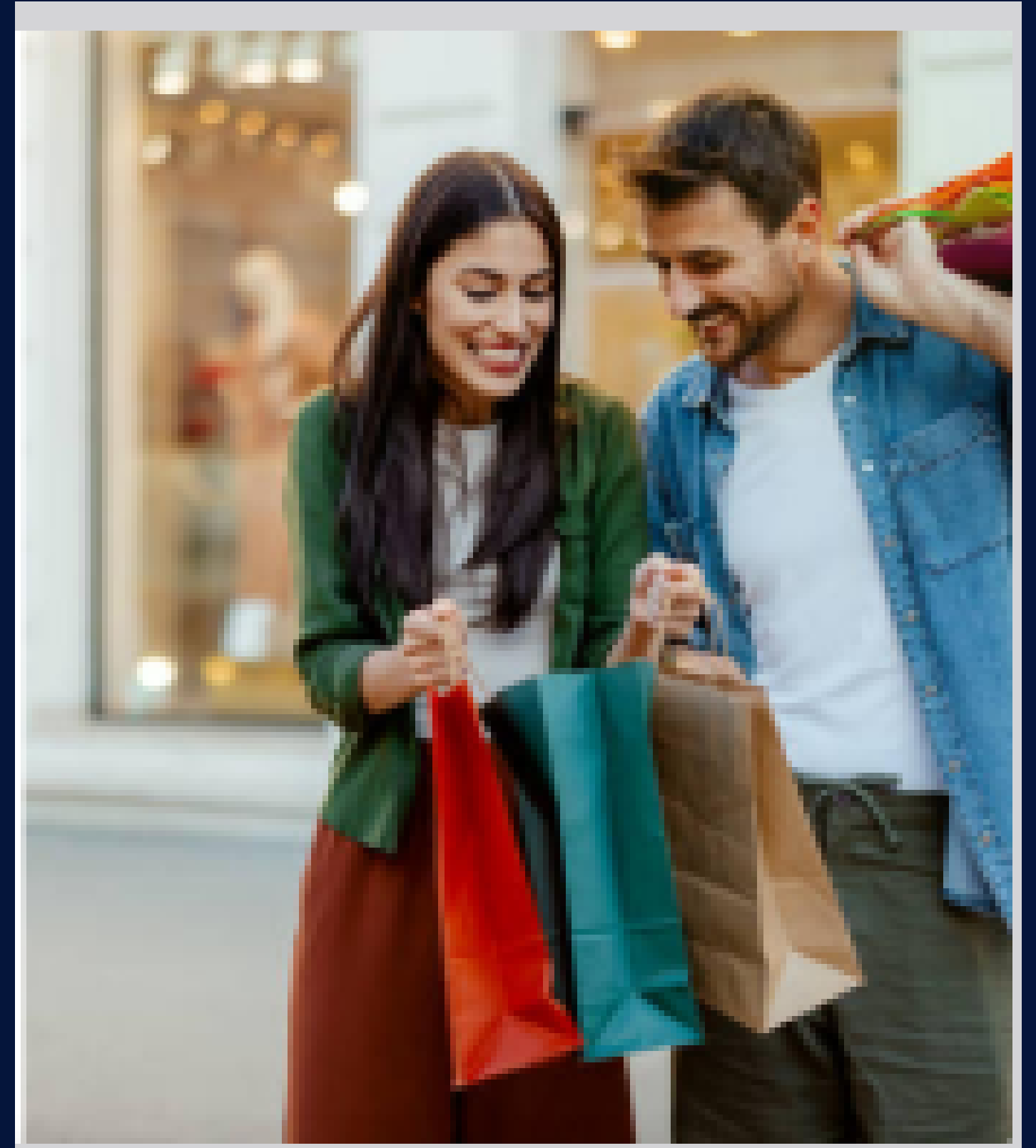
TARGET CHANNEL: DIRECT EMAIL & PRIVATE INVITES



Persona: Carefree Young

- **Profile:** Students and young professionals. Low income but highly brand-conscious and trend-driven.
- **Strategy:** Social media influencer campaigns, fast-fashion discounts, and trendy cafe vouchers.

TARGET CHANNEL: INSTAGRAM & TIKTOK ADS



Persona: Budget Shoppers

- **Profile:** Individuals looking for maximum value. Highly sensitive to price changes and clearance events.
- **Strategy:** Flash sales, BOGO (Buy One Get One) deals, and mega clearance event alerts.

TARGET CHANNEL: PUBLIC ANNOUNCEMENTS & SMS

RETAIL MERCHANDISING SIGNS LLC

YOUR RELIABLE SOURCE FOR SIGNS, BANNERS AND TAGS

Big Tags, 8 1/2" x 11"
Many more sizes
and designs on Website

SALE

WAS \$

NOW \$

SALE

WAS \$

NOW \$

CLEARANCE

WAS \$

NOW \$

BLOW OUT PRICE!

\$

Nobody BEATS THIS PRICE

GUARANTEED

SCRATCH & DENT SALE

WAS \$

NOW \$

SALE

WAS \$

NOW \$

Furniture SALE

WAS \$

NOW \$

SOFA & LOVESEAT SPECIAL!

WAS \$

NOW \$

MATTRESS SALE

TWIN

FULL

QUEEN

KING

OUR EVERYDAY LOW MATTRESS PRICE

TWIN

FULL

QUEEN

KING

Floor SAMPLE

WAS \$

NOW \$

BEDROOM GROUP SALE

WAS \$

NOW \$

LIVING ROOM GROUP SALE

WAS \$

NOW \$

DINING ROOM GROUP SALE

WAS \$

NOW \$

DINETTE SALE

WAS \$

NOW \$

ENTIRE GROUP SALE

WAS \$

NOW \$

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Statistical Model Evaluation

Mathematical Soundness

The Silhouette Score of 0.408 confirms that our 3D clusters are well-separated and internally dense. This is a strong result for a real-world dataset with behavioral variance.

By cross-validating with Gender, we ensured that the segments represent stable demographic personas rather than random noise.

0.408

Silhouette Score

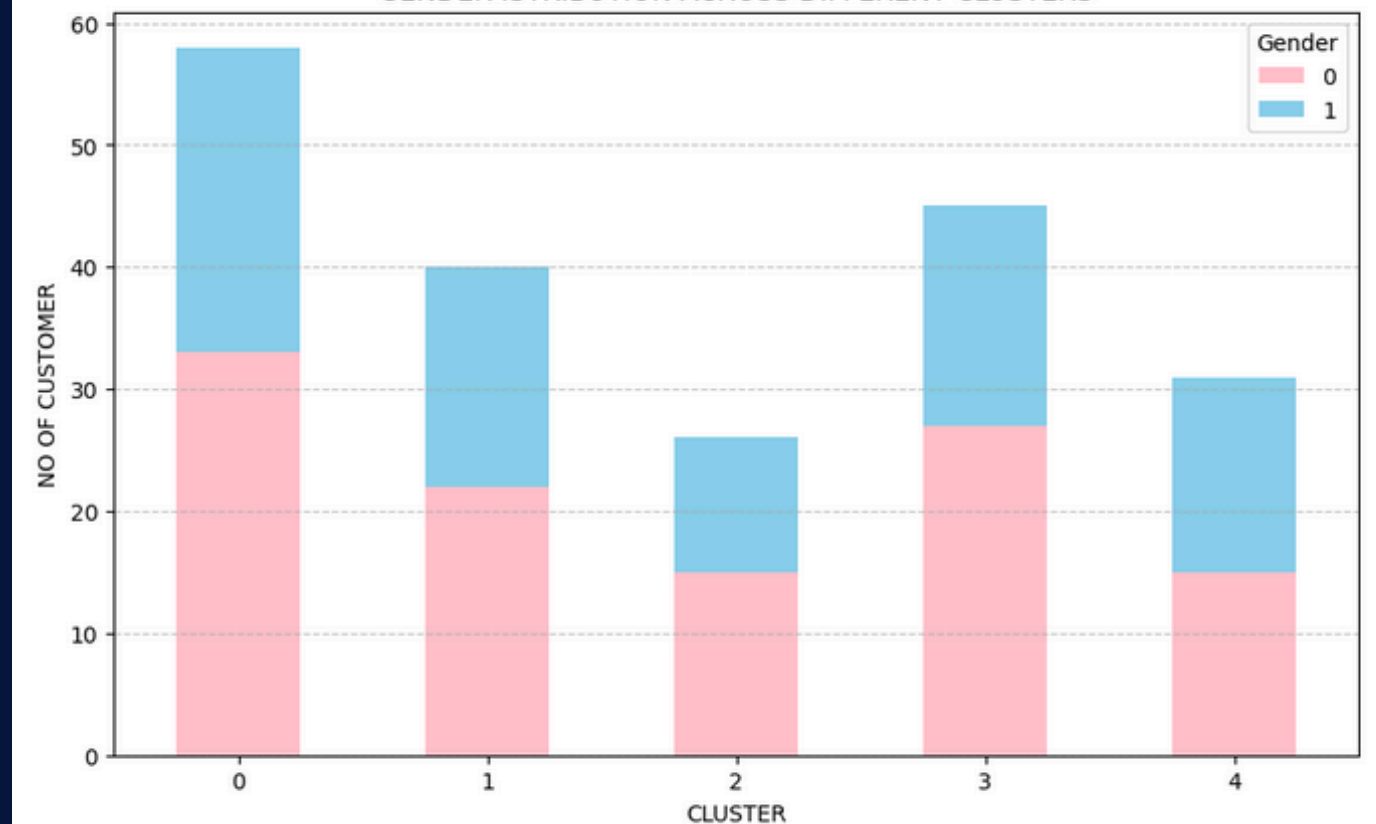
Gender Analysis Within Clusters

- Gender distribution varies across different customer clusters.
- Female customers dominate most clusters, especially among conservative and sensible spenders.
- The VIP customer segment contains a nearly balanced ratio of males and females.
- Budget shopper clusters show a slightly higher participation of male customers.
- This analysis helps businesses design gender-focused marketing and promotional strategies.

GENDER DISTRIBUTION TABLE:

Gender	0	1
Cluster		
0	33	25
1	22	18
2	15	11
3	27	18
4	15	16

GENDER ISTRIBUTION ACROSS DIFFERENT CLUSTERS



Comparative Analysis

Cluster ID	Persona Name	Dominant Gender	Avg. Spending Score
0	Sensible Spenders	Female	45-55
1	VIP Stars	Balanced	80-95
2	Trend Setters	Female	60-80
3	Conservative	Male	10-30
4	Budget Shoppers	Male	05-35

Actionable Deployment

Real-Time Decision Engine

I developed a deployment simulation function that enables the mall's POS system to instantly categorize new members:

- **Input: Age, Income, Spending Score**
- **Process: Dynamic Scaling & Prediction**
- **Output: Instant Marketing Action Strategy**

```
[16]: # 1. Freshly fit the final model with 5 clusters to clear old memory
final_kmeans = KMeans(n_clusters=5, init='k-means++', random_state=42)
final_kmeans.fit(standardscaled)

# 2. Optimized & Short Deployment Function
def smart_predict(age, income, spending_score):
    # Creating a DataFrame to avoid the feature names warning
    new_customer = pd.DataFrame([[age, income, spending_score]],
                                columns=['Age', 'Annual Income (k$)', 'Spending Score (1-100)'])

    # Scale and Predict using correct variables
    scaled_data = standardscaler.transform(new_customer)
    predicted_cluster = final_kmeans.predict(scaled_data)[0]

    # Simple strategy mapping
    strategies = {
        0: "Sensible Middle-Aged Spenders -> Send standard loyalty program perks.",
        1: "The VIP Customers (Stars) -> Send premium luxury brand rewards!",
        2: "Carefree Young Spenders -> Send trendy fashion alerts & cafe coupons.",
        3: "Conservative Spenders -> Offer value bundle promotions.",
        4: "Budget Shoppers -> Target with mega clearance sale alerts."
    }

    print(f"Input -> Age: {age} | Income: ${income}k | Spend Score: {spending_score}")
    print(f"Result -> Assigned Cluster {predicted_cluster}: {strategies[predicted_cluster]}\n")

# --- TEST THE SYSTEM ---
smart_predict(32, 120, 90) # Rich & High Spender
smart_predict(21, 18, 45) # Young & Budget Spender

Input -> Age: 32 | Income: $120k | Spend Score: 90
Result -> Assigned Cluster 1: The VIP Customers (Stars) -> Send premium luxury brand rewards!

Input -> Age: 21 | Income: $18k | Spend Score: 45
Result -> Assigned Cluster 2: Carefree Young Spenders -> Send trendy fashion alerts & cafe coupons.
```

Conclusion

- **Successful Pipeline:** We successfully built an end-to-end Unsupervised Machine Learning pipeline—moving from data preprocessing and scaling to 3D cluster visualization and dynamic deployment simulation.
- **Clear Segments Discovered:** The K-Means algorithm successfully grouped the mall customers into 5 distinct behavioral personas based on their Age, Income, and Spending habits.
- **Robust Validation:** The model achieved a Silhouette Score of 0.4085, which statistically proves that the cluster boundaries are well-separated, reliable, and meaningful for real-world marketing applications.



Recommendation

Strategic Recommendations

1. **Targeted Marketing Reallocation:** The mall management should shift away from generic mass marketing and divide the budget based on customer segments.
2. **High-ROI Focus Groups:** > * **Cluster 1 (VIP Stars):** Allocate premium budget for personalized luxury rewards, private lounge access, and exclusive brand previews.
 - **Cluster 2 (Carefree Young):** Deploy budget toward social media campaigns (Instagram/TikTok), trend-driven pop-ups, and cafe discounts to capture impulse buying behavior.
3. **Next-Phase Implementation:** Run an A/B Testing campaign over the next 90 days to compare the revenue lift generated by these targeted strategies against the old traditional marketing methods.



Thank You

THANK YOU FOR YOUR ATTENTION

